Where Can Jaguarundi Live?

Bayesian habitat suitability modeling in Ecuador with ArcGIS Pro and R

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Overview

This study applied a Bayesian habitat suitability modeling approach to predict the spatial distribution of jaguarundi in Ecuador while explicitly incorporating uncertainty into the analysis. The workflow combined GIS-based environmental data preparation in ArcGIS Pro with Bayesian logistic regression modeling in R, resulting in a set of probabilistic maps that provide both predictions and confidence measures.

Preparing the Predictors

The process began with the preparation of environmental predictor variables, which are known to influence jaguarundi habitat preferences. Elevation, landcover, and proximity to water were selected as key predictors. Landcover data were reclassified into suitability values based on ecological relevance, with dense forest and water-associated classes assigned higher suitability scores, while urban and bare areas were considered unsuitable. Additionally, a distance-to-water raster was generated and transformed using an exponential decay function to reflect the ecological reality that jaguarundi is more likely to occur near water sources. This transformation ensured that proximity to water had a non-linear influence on habitat suitability, which is more realistic than treating distance as a linear variable. Raster values for these predictors were then extracted to species presence–absence points, forming the dataset used for modeling.

Bayesian Model

In R, the predictors were standardized to ensure comparability and to improve model stability. A Bayesian logistic regression model was then fitted using these variables, where the response variable represented species presence or absence. Unlike traditional regression approaches, this model incorporated prior distributions and generated posterior distributions for each parameter, allowing uncertainty to be quantified. Model diagnostics, including trace plots and posterior predictive checks, were used to confirm that the model converged properly and adequately represented the observed data. This step ensured that the resulting predictions were statistically reliable and not influenced by poor model performance.

Predicted Suitability and Uncertainty

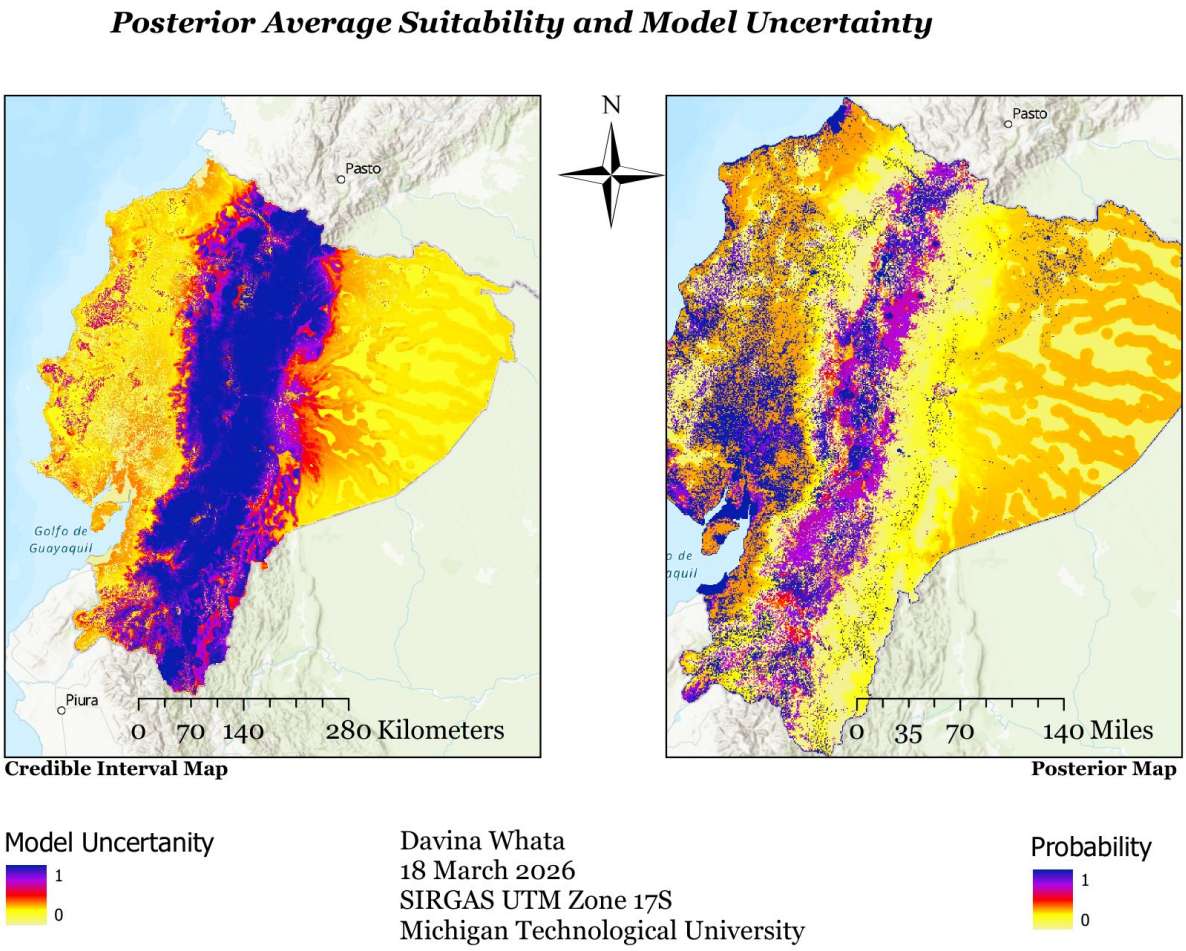


Figure 1. Model uncertainty (credible interval width, left) and posterior mean habitat suitability (right).

Using the fitted model, spatial predictions were generated by applying the posterior mean coefficients to the standardized raster layers. This produced the posterior mean suitability map, which represents the average predicted probability of jaguarundi presence across Ecuador. The results show a clear concentration of high suitability along the Andean corridor and western regions, forming a continuous band of favorable habitat. This pattern reflects the combined influence of suitable landcover, moderate elevation, and proximity to water, all of which are more prevalent in these areas. In contrast, lower suitability values dominate the eastern regions, suggesting less favorable environmental conditions or weaker predictor influence.

To complement these predictions, uncertainty was quantified through the credible interval width raster, which measures the spread of predicted probabilities across posterior simulations. Areas with high uncertainty are particularly evident in transitional zones between the Andes and surrounding lowlands. These regions likely represent areas where environmental conditions are more variable or where fewer observation points were available, leading to less confident predictions. Conversely, low-uncertainty areas correspond closely with regions of high predicted suitability, indicating that the model is both confident and consistent in these locations.

Most Likely Habitat

Areas of High Habitat Suitability (Top 25% Probability)

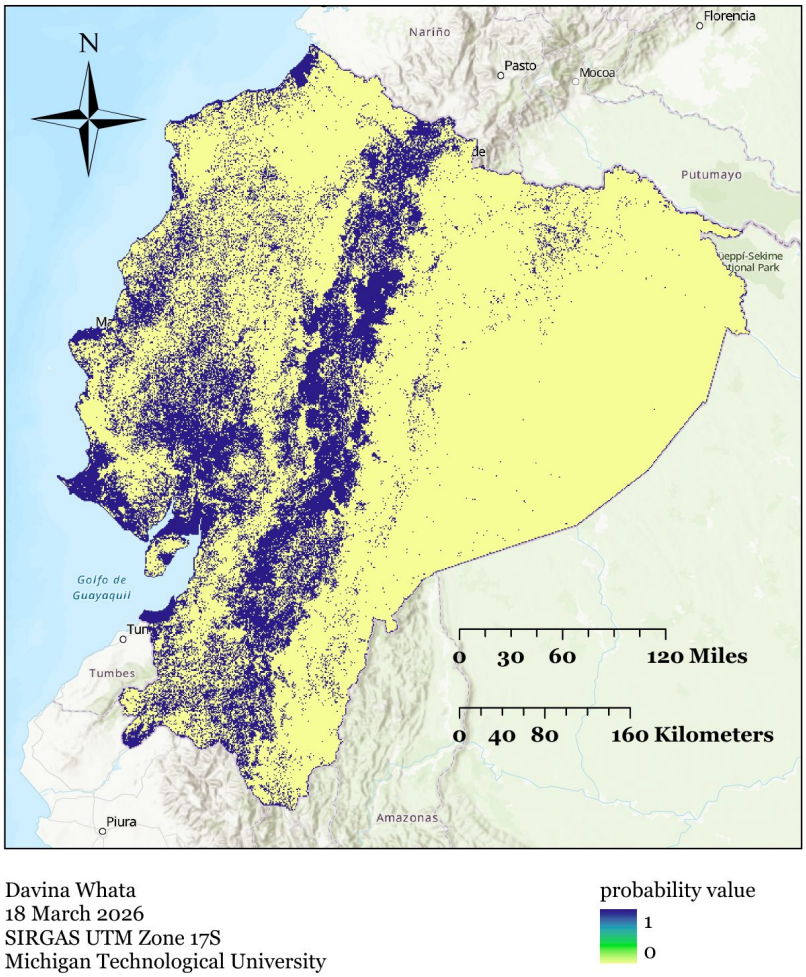


Figure 2. Areas of high habitat suitability (top 25% probability).

The analysis further identified areas exceeding the 75th percentile probability threshold, highlighting the most likely suitable habitats. These high-probability areas largely overlap with the Andean and western regions, reinforcing the importance of these zones for jaguarundi conservation. This step translates continuous probability values into a more interpretable format, making it easier to identify priority areas for management.

Robust vs. Risky Habitat

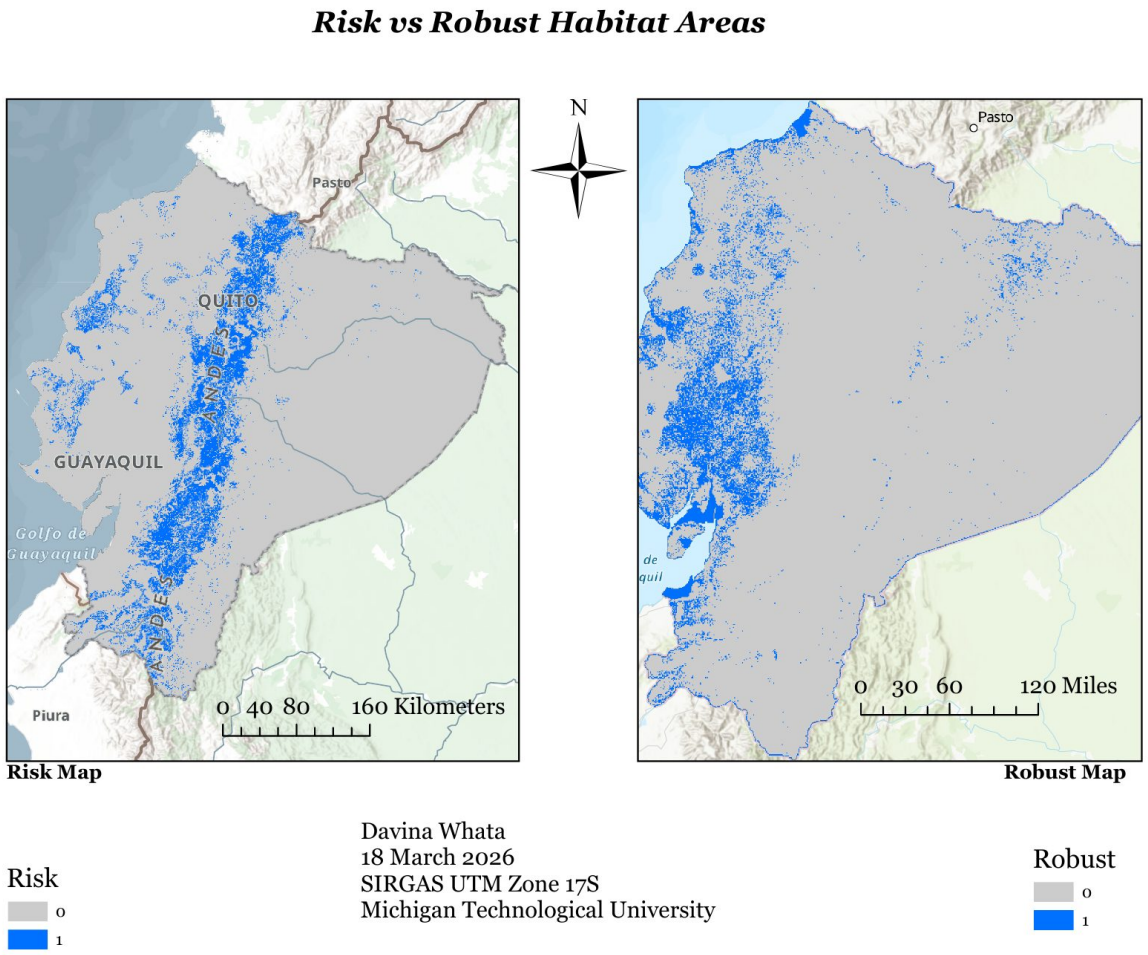


Figure 3. Risky areas (high suitability, high uncertainty, left) and robust areas (high suitability, low uncertainty, right).

A key strength of the Bayesian approach is the ability to integrate prediction and uncertainty to classify robust and risky habitat areas. Robust areas defined by high suitability and low uncertainty are concentrated along the Andes and western Ecuador, indicating regions where conservation actions can be implemented with high confidence. In contrast, risky areas where high suitability coincides with high uncertainty are more scattered and often occur at the margins of suitable habitat zones. These areas highlight where additional data collection is necessary before making definitive management decisions. This distinction is particularly valuable because it prevents overconfidence in model outputs and encourages a more cautious, evidence-based approach to conservation planning.

Limitations and Conclusion

Overall, the results demonstrate that the Bayesian modeling framework provides a more comprehensive understanding of habitat suitability compared to traditional deterministic methods. By incorporating uncertainty directly into the modeling process, it allows for more informed decision-making and highlights areas where predictions are both strong and reliable versus those that require further investigation. However, the model is not without limitations. The reliance on a limited set of predictors may oversimplify the ecological complexity of jaguarundi habitat, and areas of high uncertainty indicate the need for additional data or improved variable selection. Despite these limitations, the approach successfully

identifies key habitat corridors and provides a robust foundation for conservation planning.

In conclusion, the integration of GIS-based environmental processing with Bayesian statistical modeling resulted in meaningful and interpretable outputs that capture both habitat suitability and prediction uncertainty. This combined approach not only identifies where jaguarundi is most likely to occur but also provides critical insight into how confident we can be in those predictions, making it a powerful tool for ecological analysis and decision-making.